Loss Functions as Noise Models

Signal Detection and Estimation Theory, Project 3

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1 Introduction

1.1 The problem

When we fit a model to data, we pick a loss function and choose the parameters that make the loss as small as possible. The two most common losses are the mean squared error (MSE) and the mean absolute error (MAE). The survey by Terven et al. [2] gives rules for choosing between them, for example that MAE is better when the data has outliers. But it doesn't explain these rules using estimation theory.

Minimizing a loss is the same as maximising likelihood. So every loss assumes some distribution for the noise. We later show in Section 2 we show that MSE assumes Gaussian noise and MAE assumes Laplace noise.

In this project we observe what happens when this assumption is wrong. For example, if we use MSE but the noise is actually Laplace, does the estimate still be correct on average? what about the variance? Intuitively we'd expect variance to be larger than that of the best possible estimator.

We measure this loss in accuracy using the Cramér–Rao lower bound (CRLB), which provides a lower bound for the variance of any unbiased estimator [1]. For our study, we find the ratio of the CRLB for the true noise, to the variance of our estimator,

$$\text{efficiency ratio} = \frac{\text{Var}(\hat{A})}{\text{CRLB}}.$$

A ratio of 1 simply means the estimator is as good as possible. A ratio of 2 means its variance is twice the CRLB, hence it needs twice as many samples to be as accurate as the best estimator.

1.2 The setup

We start with the simplest case, which is estimating a DC level in noise, as in Kay. We observe N samples

$$x[n] = A + w[n], \qquad n = 0, 1, \ldots, N-1, \tag{1.1}$$

where $A \in \mathbb{R}$ is the unknown constant and $w[n]$ is the noise. The noise samples $w[n]$ are i.i.d. with zero mean and an assumed density $p_w(w)$ that is symmetric about zero, so $p_w(-w) = p_w(w)$. We want to estimate the parameter A from the samples.

Stacking the samples and the noise into vectors,

$$\mathbf{x} = [x[0], x[1], \ldots, x[N-1]]^\top, \qquad \mathbf{w} = [w[0], w[1], \ldots, w[N-1]]^\top,$$

the model becomes

$$\mathbf{x} = A\mathbf{1} + \mathbf{w},$$

where $\mathbf{1} = [1, 1, \ldots, 1]^\top$ is the column vector of N ones. This is a special case of linear regression, $\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \mathbf{w}$, with $\mathbf{X} = \mathbf{1}$ and $\boldsymbol{\theta} = A$. Here $\mathbf{X}^\top\mathbf{X} = \mathbf{1}^\top\mathbf{1} = N$, so the $1/N$ in our results becomes $(\mathbf{X}^\top\mathbf{X})^{-1}$ for the general regression model, which we study later.

2 Losses as noise assumptions

In this section we derive the MSE and MAE losses. In both cases we write the likelihood of the samples for an assumed noise distribution and find the A that maximises it.

From (1.1), $w[n] = x[n] - A$, so each sample $x[n]$ has density $p_w(x[n] - A)$. The samples are i.i.d., so the likelihood is the product of these densities,

$$p(\mathbf{x}; A) = \prod_{n=0}^{N-1} p_w(x[n] - A). \tag{2.1}$$

The maximum likelihood estimator (MLE) is the value of A that makes this as large as possible for the observed samples. It is easier to work with the log of the likelihood. Since the log is an increasing function, the maximum is at the same A, so

$$\hat{A} = \arg\max_A p(\mathbf{x}; A) = \arg\max_A \ln p(\mathbf{x}; A) = \arg\max_A \sum_{n=0}^{N-1} \ln p_w(x[n] - A).$$

2.1 Gaussian noise gives squared error

Let the noise be Gaussian with zero mean and variance σ^2,

$$w[n] \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \qquad p_w(w) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{w^2}{2\sigma^2}\right).$$

Then $x[n] \sim \mathcal{N}(A, \sigma^2)$, and from (2.1) the likelihood is

$$\begin{aligned} p(\mathbf{x}; A) &= \prod_{n=0}^{N-1} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x[n] - A)^2}{2\sigma^2}\right) \\ &= \frac{1}{(2\pi\sigma^2)^{N/2}} \exp\left(-\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)^2\right). \end{aligned}$$

Taking the log gives

$$\ln p(\mathbf{x}; A) = -\frac{N}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)^2.$$

The first term does not depend on A. The second term has a minus sign, so maximising $\ln p(\mathbf{x}; A)$ is the same as minimising

$$J(A) = \sum_{n=0}^{N-1} (x[n] - A)^2.$$

This is the squared error loss. So minimising MSE is the same as finding the MLE under Gaussian noise.

Setting the derivative of J to zero,

$$\begin{aligned} \frac{dJ}{dA} &= -2 \sum_{n=0}^{N-1} (x[n] - A) = 0 \\ &\sum_{n=0}^{N-1} x[n] - NA = 0. \end{aligned}$$

We thus obtain

$$\boxed{\hat{A} = \frac{1}{N} \sum_{n=0}^{N-1} x[n] = \bar{x}}$$

So under Gaussian noise, the MLE of A is the sample mean.

2.2 Laplace noise gives absolute error

Now let the noise be Laplace with zero mean and a scale parameter $b > 0$,

$$w[n] \overset{\text{i.i.d.}}{\sim} \text{Laplace}(0, b), \qquad p_w(w) = \frac{1}{2b} \exp\left(-\frac{|w|}{b}\right).$$

This density is also symmetric about zero, but its tails are heavier because the Gaussian density falls off like e^{-w^2}, while the Laplace density only falls off like $e^{-|w|}$. So large noise values are more likely with Laplace noise, thus the data might tend to have a few samples far away from the rest.

We follow the same steps as in the Gaussian case. From (2.1) the likelihood is

$$\begin{aligned} p(\mathbf{x}; A) &= \prod_{n=0}^{N-1} \frac{1}{2b} \exp\left(-\frac{|x[n] - A|}{b}\right) \\ &= \frac{1}{(2b)^N} \exp\left(-\frac{1}{b} \sum_{n=0}^{N-1} |x[n] - A|\right), \end{aligned}$$

and its log is

$$\ln p(\mathbf{x}; A) = -N \ln(2b) - \frac{1}{b} \sum_{n=0}^{N-1} |x[n] - A|.$$

Again the first term does not depend on A, so maximising $\ln p(\mathbf{x}; A)$ is the same as minimising

$$J(A) = \sum_{n=0}^{N-1} |x[n] - A|.$$

This is the absolute error loss. So minimising MAE is the same as finding the MLE under Laplace noise.

Here we cannot just set the derivative to zero, because $|x[n] - A|$ has a sharp edge at $A = x[n]$ where its non-differentiable. Everywhere else the derivative is

$$\frac{dJ}{dA} = -\sum_{n=0}^{N-1} \text{sign}(x[n] - A),$$

Hence, each sample above A adds -1 and each sample below A adds $+1$. So the derivative is the number of samples below A minus the number of samples above A.

If more samples lie above A than below it, the derivative is negative and J decreases as A increases. Once more samples lie below A, the derivative is positive and J increases. As A moves from left to right it passes the samples one by one, so the derivative goes up in steps, from $-N$ to $+N$. So J is smallest where the derivative changes sign, which is where half the samples lie on each side of A. For odd N the derivative never equals zero. It jumps from -1 to $+1$ at the middle sample, so the minimum is at this corner of J, which is the sample median,

$$\boxed{\hat{A} = \text{median}\big(x[0], x[1], \ldots, x[N-1]\big)}$$

For example, with the samples 1, 2 and 10, the derivative is $-3, -1, +1, +3$ as A passes each sample, so J is smallest at $A = 2$. Thus, under Laplace noise, the MLE of A is the sample median.

To summarise, MSE assumes Gaussian noise and gives the mean, and MAE assumes Laplace noise and gives the median!

3 The Cramér–Rao lower bound

The CRLB gives a lower bound on the variance of any unbiased estimator of A. From Kay [1, Theorem 3.1], if the likelihood satisfies the regularity condition

$$E\left[\frac{\partial \ln p(\mathbf{x}; A)}{\partial A}\right] = 0 \quad \text{for all } A,$$

then every unbiased estimator $\hat{A}$ satisfies

$$\mathrm{Var}(\hat{A}) \geq \frac{1}{I(A)}, \qquad I(A) = E\left[\left(\frac{\partial \ln p(\mathbf{x}; A)}{\partial A}\right)^2\right], \tag{3.1}$$

where $I(A)$ is the Fisher information. When $\ln p(\mathbf{x}; A)$ is twice differentiable, the Fisher information can also be written as

$$I(A) = -E\left[\frac{\partial^2 \ln p(\mathbf{x}; A)}{\partial A^2}\right]. \tag{3.2}$$

From (2.1), the log-likelihood is a sum over the samples, so its derivative is also a sum,

$$\frac{\partial \ln p(\mathbf{x}; A)}{\partial A} = \sum_{n=0}^{N-1} s(x[n]; A), \qquad s(x; A) = \frac{\partial}{\partial A} \ln p_w(x - A).$$

Each term $s(x[n]; A)$ depends on only one sample, so the terms are independent, and each has zero mean by the regularity condition. The variance of a sum of independent zero-mean terms is the sum of their variances, so

$$I(A) = E\left[\left(\sum_{n=0}^{N-1} s(x[n]; A)\right)^2\right] = \sum_{n=0}^{N-1} E\left[s(x[n]; A)^2\right] = N\, i(A), \qquad i(A) = E\left[s(x; A)^2\right]. \tag{3.3}$$

So we only need $i(A)$, the Fisher information of a single sample.

3.1 CRLB for Gaussian noise

For Gaussian noise,

$$\ln p_w(x - A) = -\frac{1}{2} \ln(2\pi\sigma^2) - \frac{(x - A)^2}{2\sigma^2}.$$

Differentiating with respect to A,

$$s(x; A) = \frac{x - A}{\sigma^2} = \frac{w}{\sigma^2}.$$

So the information in one sample is

$$i(A) = \frac{E[w^2]}{\sigma^4} = \frac{\sigma^2}{\sigma^4} = \frac{1}{\sigma^2}.$$

The second form also (3.2) gives the same answer, since $\partial^2 \ln p_w(x - A)/\partial A^2 = -1/\sigma^2$ for every x. Thus using the i.i.d additivity property of Fisher information, $I(A) = N/\sigma^2$, and we obtain

$$\boxed{\mathrm{CRLB}_{\mathrm{Gauss}} = \frac{\sigma^2}{N}}$$

3.2 CRLB for Laplace noise

For Laplace noise,

$$\ln p_w(x - A) = -\ln(2b) - \frac{|x - A|}{b}.$$

For $x \neq A$, the derivative with respect to A is

$$s(x; A) = \frac{\text{sign}(x - A)}{b} = \frac{\text{sign}(w)}{b}.$$

The sign is either $+1$ or -1, so $s(x; A)^2 = 1/b^2$ for every sample. Then

$$i(A) = E\left[\frac{1}{b^2}\right] = \frac{1}{b^2},$$

so $I(A) = N/b^2$, and we thus obtain

$$\boxed{\text{CRLB}_{\text{Laplace}} = \frac{b^2}{N}}$$

The regularity condition holds, because the Laplace density is symmetric and so

$$E[s(x; A)] = \frac{1}{b}\big(P(w > 0) - P(w < 0)\big) = \frac{1}{b}\left(\frac{1}{2} - \frac{1}{2}\right) = 0.$$

The derivative does not exist at $x = A$, but this single point has probability zero, so it does not change any of the expectations.

Note that we could only use the first form (3.1) of Fisher information here. The second derivative of $|x - A|$ is zero everywhere except at $x = A$, so the second form (3.2) would give $i(A) = 0$ and an infinite bound, which is wrong! As mentioned above, the second form needs $\ln p_w$ to be twice differentiable, and it's not the case here with the Laplacian noise!
Kay [1] also states that an unbiased estimator reaches the CRLB only if we can write an expression of the type:

$$\frac{\partial \ln p(\mathbf{x}; A)}{\partial A} = I(A)\big(g(\mathbf{x}) - A\big)$$

for some function g. Under Gaussian noise this holds with $g(\mathbf{x}) = \bar{x}$. Under Laplace noise the derivative is $\frac{1}{b}\sum_n \text{sign}(x[n] - A)$, which is a step function of A and cannot be written in this form. So for a fixed N, no unbiased estimator reaches the Laplace bound exactly. In Section 4.2 we see that the median reaches it as N grows!

4 Estimators under the wrong noise

We now find the variance of the sample mean and the sample median under both noise types, and compare each with the CRLB from Section 3.

4.1 The sample mean

The sample mean is unbiased for any noise with zero mean,

$$E[\bar{x}] = \frac{1}{N}\sum_{n=0}^{N-1} E\big[A + w[n]\big] = A.$$

The samples are independent, so

$$\mathrm{Var}(\bar{x}) = \frac{1}{N^2}\sum_{n=0}^{N-1}\mathrm{Var}(w[n]) = \frac{\mathrm{Var}(w)}{N}. \tag{4.1}$$

This only uses the variance of the noise so far, without considering the shape of a distribution.

4.1.1 Under Gaussian Noise

Under Gaussian noise, $\mathrm{Var}(w) = \sigma^2$, so $\mathrm{Var}(\bar{x}) = \sigma^2/N$. This equals the CRLB, so the sample mean is efficient under Gaussian noise.

4.1.2 Under Laplacian Noise

Under Laplace noise, the noise has zero mean, so $\mathrm{Var}(w) = E[w^2] - (E[w])^2 = E[w^2]$. Writing out the expectation,

$$E[w^2] = \int_{-\infty}^{\infty} w^2\,\frac{1}{2b}e^{-|w|/b}\,dw.$$

$$E[w^2] = 2\cdot\frac{1}{2b}\int_0^{\infty} w^2 e^{-w/b}\,dw = \frac{1}{b}\int_0^{\infty} w^2 e^{-w/b}\,dw.$$

We find the integral using integration by parts twice. Let

$$I_k = \int_0^{\infty} w^k e^{-w/b}\,dw.$$

Using integration by parts:

$$\int_0^{\infty} u\,dv = \Big[uv\Big]_0^{\infty} - \int_0^{\infty} v\,du.$$

For I_2 we take $u = w^2$ and $dv = e^{-w/b}\,dw$, so $du = 2w\,dw$ and $v = -b\,e^{-w/b}$. Then

$$\begin{aligned} I_2 &= \Big[-b\,w^2 e^{-w/b}\Big]_0^{\infty} + 2b\int_0^{\infty} w\,e^{-w/b}\,dw \\ &= 0 + 2b\,I_1. \end{aligned}$$

The boundary term is zero, because at $w = 0$ the factor w^2 is zero, and as $w \to \infty$ the exponential $e^{-w/b}$ goes to zero faster than w^2 grows. In the same way, for I_1 we take $u = w$ and $dv = e^{-w/b}\,dw$,

$$\begin{aligned} I_1 &= \Big[-b\,w\,e^{-w/b}\Big]_0^{\infty} + b\int_0^{\infty} e^{-w/b}\,dw \\ &= 0 + b\Big[-b\,e^{-w/b}\Big]_0^{\infty} \\ &= b\cdot b = b^2. \end{aligned}$$

Putting these together, $I_2 = 2b\cdot b^2 = 2b^3$, and so

$$\begin{aligned} \mathrm{Var}(w) = E[w^2] &= \frac{1}{b}\,I_2 \\ &= 2b^2. \end{aligned}$$

Now from (4.1),

$$\boxed{\mathrm{Var}(\bar{x})_{\mathrm{Laplace}} = \frac{2b^2}{N} = 2 \times \mathrm{CRLB}_{\mathrm{Laplace}}}$$

This holds for every N, not only for large N! The mean is still correct on average, but its variance is twice the CRLB, so it needs twice as many samples to be as accurate as the best estimator! The intuitive reason is that every sample gets the same weight $1/N$, including the few large samples that Laplace noise produces, and these pull the mean away from A.

4.2 The sample median

The noise is symmetric about zero, so the samples are symmetric about A, and so is the distribution of the median. So the median is also unbiased. Its exact variance has no simple formula, but for large N the sample median is approximately Gaussian with variance [3, Ch. 21]

$$\mathrm{Var}(\mathrm{median}) \approx \frac{1}{4N\, p_w(0)^2}. \tag{4.2}$$

The median only depends on which samples lie above and below A, so its accuracy depends on how many samples fall close to A. This is set by the height of the noise density at the centre, $p_w(0)$. A taller peak puts more samples near A and gives a more accurate median.

For Laplace noise, $p_w(0) = 1/(2b)$, so from (4.2)

$$\boxed{\mathrm{Var}(\mathrm{median})_{\mathrm{Laplace}} \approx \frac{1}{4N \cdot \frac{1}{4b^2}} = \frac{b^2}{N} = \mathrm{CRLB}_{\mathrm{Laplace}}}$$

So for large N the median reaches the CRLB under Laplace noise. It is asymptotically efficient.

For Gaussian noise, $p_w(0) = 1/\sqrt{2\pi\sigma^2}$, so

$$\boxed{\mathrm{Var}(\mathrm{median})_{\mathrm{Gauss}} \approx \frac{1}{4N \cdot \frac{1}{2\pi\sigma^2}} = \frac{\pi}{2} \cdot \frac{\sigma^2}{N} \approx 1.57 \times \mathrm{CRLB}_{\mathrm{Gauss}}}$$

Under Gaussian noise, the distance of each sample from A carries information. The median throws this away, since it only looks at the order of the samples, and this costs about 57% more variance than the mean.

4.3 Predicted efficiency ratios

Dividing each variance by the CRLB for the true noise gives the efficiency ratios in Table 1.

Estimator (loss)	Gaussian noise	Laplace noise
Sample mean (MSE)	1	2
Sample median (MAE)	$\pi/2 \approx 1.57$	1

Table 1: Predicted efficiency ratio $\mathrm{Var}(\hat{A})/\mathrm{CRLB}$ for each estimator and each true noise.

Each estimator is efficient under the noise it assumes and loses efficiency under the other one. The values for the mean are exact for every N, while the values for the median hold only for large N. We check these values by Monte Carlo simulation in Section 5.

5 Simulation results

In this section we check the results of Section 4 by simulation, and then look at two noise types where the theory so far does not directly apply. All the code is in the `sim/` folder of the project, with one script per simulation.

Every simulation follows the same Monte Carlo method. We fix the true value $A = 1$ and repeat the experiment $R = 5000$ times. In each trial we draw N fresh noise samples, form $x[n] = A + w[n]$, and compute both the sample mean and the sample median. This gives 5000 values of each estimator. Their average estimates $E[\hat{A}]$, and their sample variance estimates $\mathrm{Var}(\hat{A})$. Dividing the variance by the CRLB gives the measured efficiency ratio.

Since we only use 5000 trials, the measured variance is itself a little noisy. For estimates that are roughly Gaussian, the relative error of a sample variance computed from R values is about $\sqrt{2/R}$, which is 2% here. So a measured ratio of 0.97 is in agreement with a predicted ratio of 1.

5.1 Gaussian and Laplace noise

We first check the predicted ratios of Table 1. We use $\sigma = 1$ for the Gaussian noise and $b = 1$ for the Laplace noise. The ratios do not depend on these values, since both the variances and the CRLBs scale in the same way. The code is in `sim/sim_A_mean_vs_median.py`.

Table 2 shows the measured ratios at $N = 100$ next to the predicted ones. All four agree with the theory. The mean has ratio 1 under Gaussian noise and 2 under Laplace noise, and the median has ratio close to $\pi/2$ under Gaussian noise.

	Gaussian noise		Laplace noise	
Estimator	Predicted	Measured	Predicted	Measured
Sample mean (MSE)	1	0.968	2	2.007
Sample median (MAE)	1.571	1.512	1	1.169

Table 2: Predicted and measured efficiency ratio $\mathrm{Var}(\hat{A})/\mathrm{CRLB}$ at $N = 100$, from 5000 trials.

The only value that is clearly away from its prediction is the median under Laplace noise, 1.169 instead of 1. This is expected, because (4.2) only holds for large N. To see this, we repeat the experiment for N from 10 to 1000. Figure 5.1 shows the result.

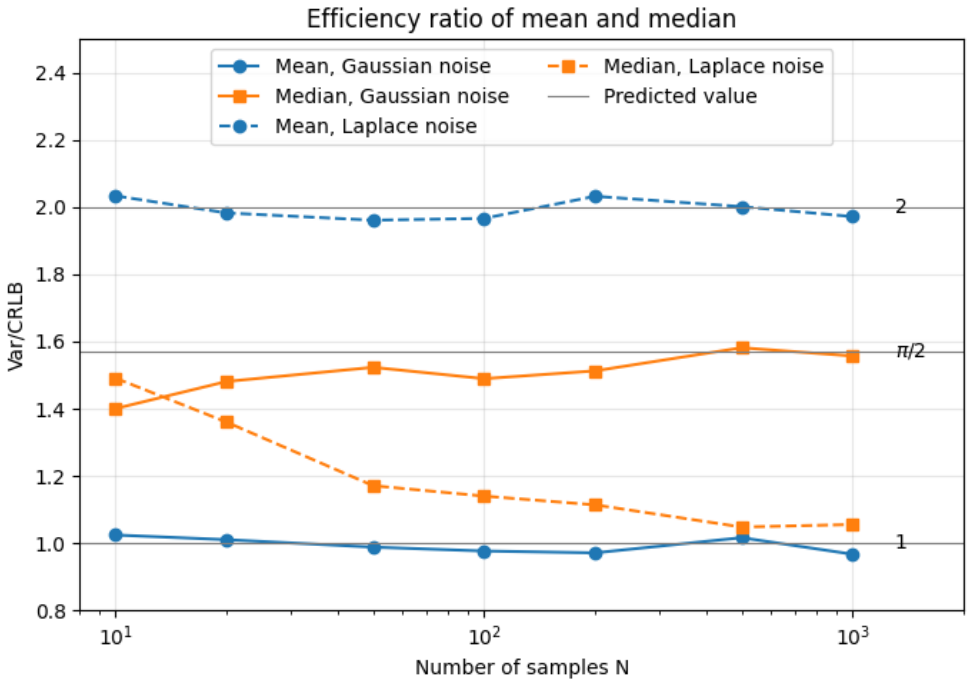


Figure 5.1: Measured efficiency ratio against the number of samples N. The grey lines are the predicted values 1, $\pi/2$ and 2.

We can observe two different behaviours in Figure 5.1. The two curves for the mean are flat, at 1 for Gaussian noise and 2 for Laplace noise, already from $N = 10$. This matches Section 4, where we found $\mathrm{Var}(\bar{x}) = \mathrm{Var}(w)/N$ exactly for every N. The two curves for the median change with N. Under Laplace noise the ratio starts at about 1.5 for $N = 10$ and slowly comes down towards 1, reaching 1.05 at $N = 1000$. Under Gaussian noise it starts at 1.4 and rises towards $\pi/2$. So the median only becomes efficient under Laplace noise once N is large, while the mean is twice as bad as the best estimator at every N.

5.2 Asymmetric noise

So far the noise has been symmetric about zero. In that case the mean and the median both aim at the same value A, and they only differ in their variance. Real noise is often skewed, with a long tail on one side. For skewed noise the mean and the median of the noise are no longer equal, and the two estimators aim at different values. In this section we show this, and we predict the value each estimator reaches before running the simulation.

What each estimator converges to. The sample mean converges to the mean of the distribution of $x[n]$, which is $A + E[w] = A$, since the noise still has zero mean. The sample median balances the samples, with half on each side, so for large N it converges to the value that splits the distribution of $x[n]$ in half. This is the median of $x[n]$, which is $A + \mathrm{median}(w)$. So

$$\bar{x} \to A, \qquad \mathrm{median}(\mathbf{x}) \to A + \mathrm{median}(w).$$

This is the same idea as in Section 2. Squared error leads to the mean and absolute error leads to the median, and when the noise is skewed these are two different numbers.

Exponential noise. Let $E \sim \text{Exp}(1)$, which has mean 1, and set $w = E - 1$ so that w has zero mean. The median m of E satisfies

$$P(E \le m) = 1 - e^{-m} = \frac{1}{2},$$

which gives $m = \ln 2$. So $\text{median}(w) = \ln 2 - 1 \approx -0.307$, and we predict

$$\bar{x} \to 1, \qquad \text{median}(\mathbf{x}) \to 1 + \ln 2 - 1 = \ln 2 \approx 0.693.$$

Lognormal noise. Let $Z \sim \mathcal{N}(0, 1)$. Then e^Z is lognormal with mean $E[e^Z] = e^{1/2}$, and we set $w = e^Z - e^{1/2}$ so that w has zero mean. Since e^z is an increasing function, the median of e^Z is e raised to the median of Z, which is $e^0 = 1$. So $\text{median}(w) = 1 - e^{1/2} \approx -0.649$, and we predict

$$\bar{x} \to 1, \qquad \text{median}(\mathbf{x}) \to 2 - e^{1/2} \approx 0.351.$$

The lognormal noise has a much longer right tail than the exponential noise, so it produces a few very large samples.

Figure 5.2 shows both noise densities next to the standard Gaussian. Both have mean zero, but most of their probability lies to the left of zero, and this is balanced by a long tail on the right. The shaded area, to the left of the median, holds exactly half of the probability. The gap between the mean and the median is what the two estimators will see.

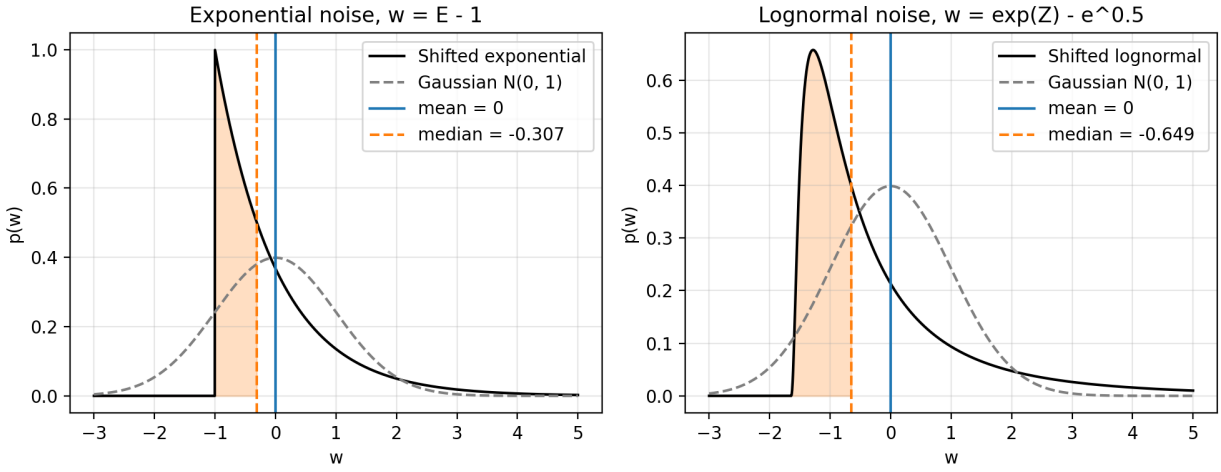


Figure 5.2: Density of the shifted exponential noise (left) and the shifted lognormal noise (right), with the standard Gaussian for comparison. The blue line is the mean and the orange dashed line is the median. The shaded area is half of the probability.

Results. The code is in `sim/sim_B_asymmetric_noise.py`. Figure 5.3 shows the histograms of the 5000 estimates at $N = 100$, and Table 3 gives their average and standard deviation.

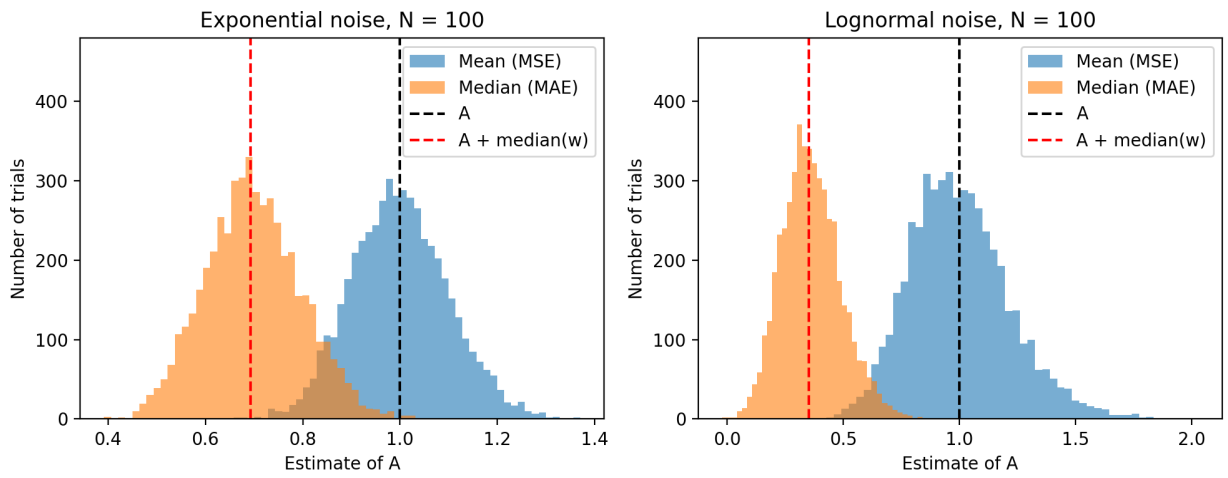


Figure 5.3: Histograms of the mean and median estimates over 5000 trials, with $N = 100$ and $A = 1$. The black line is A and the red line is $A + \text{median}(w)$.

Noise	Estimator	Predicted limit	Average	Predicted std	Measured std
Exponential	Mean	1	0.999	0.100	0.099
	Median	0.693	0.698	0.100	0.098
Lognormal	Mean	1	1.001	0.216	0.215
	Median	0.351	0.363	0.125	0.127

Table 3: Average and standard deviation of the estimates at $N = 100$, from 5000 trials.

Each estimator lands where we predicted. The mean is centred on $A = 1$ for both noises, and the median is centred on $A + \text{median}(w)$, which is 0.693 for exponential noise and 0.351 for lognormal noise. The two histograms barely overlap, so the two losses give clearly different answers from the same data.

The predicted standard deviations in Table 3 come from the same formulas as before. For the mean we use (4.1), with $\text{Var}(w) = 1$ for exponential noise and $\text{Var}(w) = (e-1)e \approx 4.67$ for lognormal noise. For the median we use (4.2), where for skewed noise $p_w(0)$ is replaced by the density of w at its median [3, Ch. 21]. This density is $1/2$ for exponential noise and $1/\sqrt{2\pi}$ for lognormal noise. Under exponential noise both estimators have the same spread. Under lognormal noise the mean has almost twice the spread of the median, because the few very large samples from the long tail pull the mean around, just as with Laplace noise.

Figure 5.4 shows the average estimate as N grows. The mean is centred on A at every N, since $E[\bar{x}] = A$ exactly. The median starts a little above its limit for small N and settles on $A + \text{median}(w)$ as N grows. So the difference between the two is not a small-sample effect. It stays no matter how much data we collect.

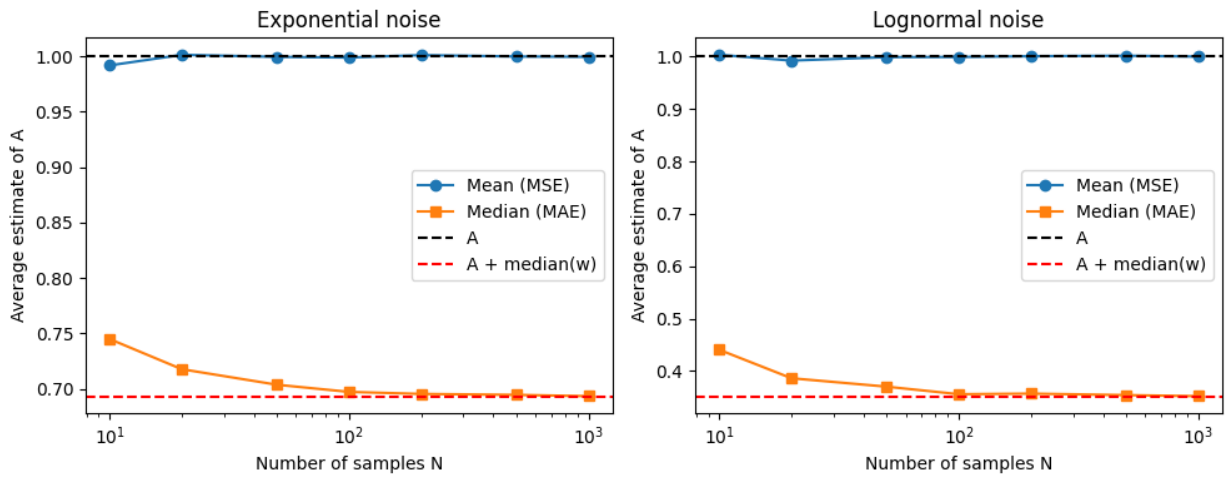


Figure 5.4: Average of the 5000 estimates against N, for exponential noise (left) and lognormal noise (right).

Note that we do not compute efficiency ratios here. The two estimators converge to different values, so comparing their variances with a single CRLB would compare estimators of two different quantities. The more important point is that the choice of loss decides which quantity we estimate. If A is meant to be the mean of the data, MSE is the right loss. If it is meant to be the typical value, which the median describes better for skewed data, MAE is the right loss. Choosing MAE because the data has outliers, as the survey suggests [2], also changes what is being estimated when the noise is skewed.

References

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